Derivations/Proofs

- Derive the equations of motion (all of them) for constant acceleration from the definitions of average acceleration and average velocity (ch 2) **using both algebraic and calculus methods.**
  Herlei die beweginsvergelykings (almal) met konstante versnelling van die definisies van gemiddelde versnelling en gemiddelde snelheid.(hfst 2) ***deur algebrais en kalkulus methodes te gebruik.***

- Show that the range of a projectile is given by $R = \frac{v_0^2}{g}\sin 2\theta_0$ from the basic equations of motion for constant acceleration. (ch 4)
  Wys dat $R = \frac{v_0^2}{g}\sin 2\theta_0$ *(die rykwydte van 'n projektiel) van die basiese vergelykings van beweging vir konstante versnelling.*

- Show that the maximum height reached by a projectile is given by $h = \frac{v_i^2}{2g}\sin^2\theta_i$ from the basic equations of motion for constant acceleration. (ch 4)
  Wys dat die maksimum hoogte wat 'n projektiel bereik gegee word deur $h = \frac{v_i^2}{2g}\sin^2\theta_i$ *. Maak gebruik van die basiese vergelykings van beweging vir konstante versnelling.*

- From first principles show that the acceleration during uniform circular motion is given by $a = \frac{v^2}{r}$.
  Van eerste beginsels, wys dat die versnelling tydens uniforme sirkelbeweging geegee word deur $a = \frac{v^2}{r}$.

- Derive the equation for the work done by a spring on a block from the basic definition of work done, $W_s = \int \vec{F}_s \cdot d\vec{r}$.
  Herlei die vergelyking vir die arbeid verrig deur 'n veer op 'n blok deur gebruik te maak van die definisie vir arbeid, $W_s = \int \vec{F}_s \cdot d\vec{r}$

- Derive the work-kinetic energy theorem starting with the definition of work done by the external net force, $W_{ext} = \int \sum \vec{F} \cdot d\vec{r}$, when the object moves horizontally in the through a displacement, $\Delta\vec{r} = \Delta x\hat{\boldsymbol{\imath}}$.
  Herlei die arbeid-kinetiese energie stelling deur te begin met die definisie van arbeid verrig deur 'n eksterne netto krag, $W_{ext} = \int \sum \vec{F} \cdot d\vec{r}$ *, wanneer die voorwerp horisontaal beweeg deur 'n verplasing,* $\Delta\vec{r} = \Delta x\hat{\boldsymbol{\imath}}$.

- Clearly show that in an elastic collision between two masses, the final velocities of the masses can be given by:

  $$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i}$$

  and

  $$v_{1f} = \frac{2m_1}{m_1 + m_2} v_{1i} + \frac{m_2 - m_1}{m_1 + m_2} v_{2i}$$

  Wys duidelik dat vir ‘’n elastiese botsing tussen twee massas in een dimensie, die finale snelhede van die twee massas word gegee deur:

  $$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i}$$

  en

  $$v_{1f} = \frac{2m_1}{m_1 + m_2} v_{1i} + \frac{m_2 - m_1}{m_1 + m_2} v_{2i}$$

- Firstly, clearly state what is understood by a linear oscillator and give an example. Then show clearly, using the basic definitions for potential and kinetic energy, that the mechanical energy E of a linear oscillator is given by $E=½kA^2$. State clearly all reasons for each step in the derivation.
  Eerstens, verduidelik wat ’n lineêre ossilator is en gee ’n voorbeeld daarvan. Dan, deur gebruik te maak van die definisies van potensiële energie en kinetiese energie, wys dat die meganiese energie E van ’n lineêre ossilator $E=½kA^2$ *is. Stel duidelike die redes vir elke stap.*

- Derive the period of a simple pendulum $T = 2\pi\sqrt{\frac{L}{g}}$.

  Herlei die vergelyking vir die periode van ’n eenvoudige slinger $T = 2\pi\sqrt{\frac{L}{g}}$.